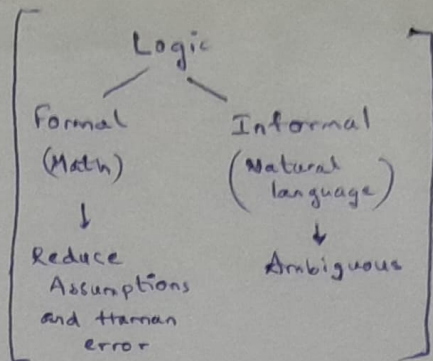


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- (i) Humanly vs. Rationally
- (ii) Act vs. Think

- Logic : Formal Methods
- Formal logic $\rightarrow$ AI
 - (i) Math logic
 - (ii) Symbolic logic
 - (iii) Propositional logic



- Reasoning : Drawing inference from knowledge
- Propositional logic $\rightarrow$ 0th order logic
- Robin's Conjecture - Proved by computer by logic
- Fallacy : Reasoning is incorrect
Conclusion may be correct

Order of logic :
degree of Quantification
that can be performed
over sets

Zeroth order logic

- Propositional logic :
 - Propositions - Declarative statements
 - Atomic Propositions - basic fact / condition
Cannot be broken down

Compound propositions - formed using logical connectives
(AND, OR, NOT) - even XOR

Statement can be either true or false.

It will rain tomorrow - considered to be proposition.

0th order : Only statements

1st order : Predicate logic

$$\forall x, x \geq 0$$

$$\exists x \in \mathbb{R}, x > 0$$

2nd order : Quantifies over sets

1st order : Quantifiers over individuals

$\hookrightarrow$ combination of AND, OR, NOT

(P) Today is Wednesday

(Q) It is raining today

(R) It will be raining tomorrow

(S) $2+7=9$

(T) $x+2=1$

n variables $\Rightarrow 2^n$ rows
in truth
table

$\rightarrow$ Connections :

AND : $\wedge$

OR : $\vee$

NOT : $\neg$ (OR) $\sim$

Implies : $\rightarrow \equiv \neg P \vee Q$

IFF : $\leftrightarrow \equiv \frac{\neg(P \oplus Q)}{\neg P \oplus Q}$

IFF : P is necessary and sufficient for Q

$(P \rightarrow Q)$ Q is necessary and sufficient for P

P & Q are equivalent.

Implies : P is sufficient for Q

$(P \rightarrow Q)$ Q is necessary for P

$\rightarrow$ Formal Proof :

$\alpha \Rightarrow \beta : \left. \begin{array}{l} \alpha : \text{True} \\ \beta : \text{True} \end{array} \right\} \text{Sets}$

e.g. $P \rightarrow Q \Leftrightarrow \neg P \vee Q$

$P \leftrightarrow Q \Leftrightarrow \neg P \oplus Q$

$P \Leftrightarrow Q$: True in P has corresponding True in Q
False in P has corresponding False in Q

• Modus Ponens : $P, P \rightarrow Q \Rightarrow Q$, i.e. $\alpha = P \wedge (P \rightarrow Q)$
 $\beta = Q$

P	Q	$P \leftrightarrow Q$	$P \rightarrow Q$	$P \vee Q$	$P \wedge Q$	$\neg P \vee Q$	$\neg P \oplus Q$	$\neg P$
0	0	1	1	0	0	1	1	1
0	1	0	1	1	0	1	0	1
1	0	0	0	1	0	0	0	0
1	1	1	1	1	1	1	1	0

Tautology : Always True

[Valid Statements] $(P \vee \neg P)$

SAT : Satisfiability

Unsatisfiability - e.g. $P \wedge \neg P$
(Fallacy)

Proof: Sequence of steps, where each step is:

- (i) Axiom / Premise
- (ii) formula deduced from previous steps
- (iii) some rule of inference

S follows from premises α to denote that the set of formulae prove the formula S .

- ① Soundness: It preserves truth (if premise true $\Rightarrow$ conclusion also true.)
- ② Completeness: Capable of proving any valid consequence. $\leftarrow$ Semantic
Every sentence/negation is provable $\leftarrow$ syntactic
- ③ Decidability: $\exists$ Decision procedure (finite) for theorem

[No logic system is syntactically complete]
 $\left\{ \begin{array}{l} \text{Zeroth Order is Semantically complete} \leftarrow 1^{st} \text{ order as well} \\ (\text{Decidable}) \leftarrow \therefore \text{syntactically complete} \quad (\text{Not decidable}) \end{array} \right\}$

Inference rules:

- (1) $P, P \rightarrow Q \Rightarrow Q$ — Modus Ponens
- (2) $P \rightarrow Q, \neg Q \Rightarrow \neg P$ — Modus Tollens
- (3) $P \rightarrow Q, Q \rightarrow R \Rightarrow P \rightarrow R$ — Hypothetical Syllogism
- (4) $P_1 \wedge P_2 \wedge \dots \wedge P_n \Rightarrow P_i$ — AND elimination
- (5) $P_1, P_2, P_3, \dots, P_n \Rightarrow P_1 \wedge P_2 \wedge \dots \wedge P_n$ — AND Introduction
- (6) $P_i \Rightarrow P_1 \vee P_2 \vee \dots \vee P_n$ — OR Introduction
- (7) $\neg \neg P \Rightarrow P$ — Double negation elimination
- (8) $P \vee Q, \neg Q \Rightarrow P$ — Unit Resolution
- (9) $P \vee Q, \neg Q \vee R \Rightarrow P \vee R$ — Resolution

e.g.

- ① $A \vee (B \rightarrow D)$
- ② $\neg C \rightarrow (D \rightarrow E)$
- ③ $A \rightarrow C$
- ④ $\neg C$

$\therefore B \rightarrow E$

- $\therefore$ ④ & ② : $D \rightarrow E$ — ⑤
 ③ & ④ : $\neg A$ — ⑥ [MP]
 ⑥ & ① : $B \rightarrow D$ — ⑦
 ⑦ & ⑤ : $B \rightarrow E$ ✓

Propositional logic
 Truth Tables
 Order of Logics
 Connectors
 Tautology
 SAT

→ Conjunctive Normal Form:

Literal: Propositional letter (P, Q, R)
its negation

Clause: Disjunction of ~~literal~~ literals

CNF: Conjunction of Clauses
e.g. $(P \vee Q \vee R) \wedge (\neg S \vee R)$

DNF: Disjunction of Conjunctions of literals

→ Disjunctive Normal Form
(e.g.) $(P \wedge Q \wedge R) \vee (\neg S \wedge R)$

Note: Every propositional logic formula can be converted to CNF or DNF.

→ Converting to CNF:

$$P \leftrightarrow Q \equiv (P \rightarrow Q) \wedge (Q \rightarrow P)$$

$$P \rightarrow Q \equiv \neg P \vee Q$$

$$\neg(P \wedge Q) \equiv \neg P \vee \neg Q$$

$$\neg(P \vee Q) \equiv \neg P \wedge \neg Q$$

$$\neg(\neg P) \equiv P$$

$$(P \wedge Q) \vee R \equiv (P \vee R) \wedge (Q \vee R)$$

$$(P \vee Q) \wedge R \equiv (P \wedge R) \vee (Q \wedge R)$$

e.g. $P \rightarrow Q$

$$\neg P \vee Q$$

$$\therefore Q \vee R$$

Prove ↑

$$(\neg P \vee Q) \wedge (P \vee R)$$

$$= (\neg P \vee Q \wedge P) \vee (\neg P \vee Q \wedge R)$$

$$= \neg P \vee Q \wedge R$$

$$(\neg P \vee Q \wedge R) \rightarrow Q \vee R$$

$$(P \wedge Q \vee R) \vee (Q \vee R) = 1$$

$$(P \wedge Q \vee R) \vee (Q \vee R) = 1$$

e.g.

$$\left\{ \begin{array}{l} P \rightarrow Q \\ Q \rightarrow R \\ R \rightarrow S \\ (P \rightarrow S) \end{array} \right.$$

$$P \rightarrow S$$

$$\therefore Q \rightarrow P$$

$$\therefore \neg Q \vee P$$

$$(P \rightarrow S)$$

$$\therefore \neg Q \vee P$$

$$\neg Q$$

Recap: Apriori
itemset

Association Rules - Identify frequently appearing patterns, AR: $X \rightarrow Y$

$$\left. \begin{array}{l} \text{Confidence} \rightarrow \frac{X \cap Y}{\text{Total}} P(Y|X) \\ \text{Support} \rightarrow \frac{X \cap Y}{\text{Total}} \text{ i.e. } P(X \cap Y) \\ \text{Lift} \rightarrow \frac{P(Y|X)}{P(Y)} \end{array} \right\} \text{Fail when dataset is skewed}$$

2^n - Brute Force - $X \cup Y$ X Use Apriori

Apriori(T, ϵ):

$L_1 \leftarrow \{\text{large singleton itemsets}\}$

$k \leftarrow 2$

while L_{k-1} is not empty:

$C_k \leftarrow \text{General_Candidates}(L_{k-1}, k)$

for transactions t in T :

$D_t \leftarrow \{c \text{ in } C_k : c \subseteq t\}$

for candidates c in D_t :

$\text{count}[c] \leftarrow \text{count}[c] + 1$

$L_k \leftarrow \{c \text{ in } C_k : \text{count}(c) \geq \epsilon\}$

$k \leftarrow k + 1$

return Union(L_k) over all k

Generate_candidates(L, k):

result $\leftarrow$ empty set()

$\forall p \in L, q \in L$ where p & q differ by 1 element

$c \leftarrow p \cup q$

if $u \in L \ \forall u \subseteq c$ where $|u| = k-1$

result.add(c)

return result

Association Rules

- Categorical
- Hierarchical
- Cyclic
- Eclat Algorithm (DFS)
- FP-Growth (Frequent- Pattern Tree)

Apriori - works well with sparse data
for Eclat Algo

- works well with dense data
for FP Growth

Data Mining - Vikram Pudi

- Classification :

$$x \in \mathbb{R}^d$$

$$x = (x_1, x_2, \dots, x_d)$$

$$y \in \{1, 2, \dots, k\} = [k]$$

$$[k] = \{1, 2, \dots, k\}$$

If $k=2 \Rightarrow$ Binary Classification

$f: X \rightarrow Y$: Classifier

$\hookrightarrow$ label set, $f: X \rightarrow \mathbb{R}$
 $\hookrightarrow$ Input set

H : Scoring function

$d=2$:

$$f(x_1, x_2) = 1 \quad \text{if } x_2 > 100$$

$$= 2 \quad \text{else}$$

pred

	P	N
P	True Positive	False Negative
N	False Positive	True Negative

True

Predicted label
True label
 $\Pr(\hat{y}_2 = y_2) \leftarrow$ Accuracy

$$[f(x) = \hat{y}_2]$$

$$= \frac{\# \text{ data points correctly classified}}{\# \text{ data points}} = \frac{TP + TN}{(TP + TN) + (FP + FN)}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{Precision} = \frac{TP}{FP + TP} = \text{PPV}$$

(Positive Predicted Value)

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{False Negative Rate} = \text{FNR} = \frac{\text{FN}}{P}$$

$$\text{True Negative Rate} = \text{TN} = \frac{\text{TN}}{N}$$

$$\text{False Positive Rate} = \text{FPR} = \frac{\text{FP}}{N}$$

$$\text{True Positive Rate} = \frac{\text{TP}}{P}$$

F_1 Score is Harmonic Mean of Precision & Recall

$$F_1 = \frac{2}{\frac{1}{\text{Recall}} + \frac{1}{\text{Precision}}}$$

→ Bayes classifier:

$$Y \in \{1, 2, 3, \dots, k\}$$

$$P(Y=k) \leftarrow \text{Prior}$$

$$P(Y=k|X=x) \leftarrow \text{Posterior}$$

$$\hat{y} \in \arg \max_k P(Y=k|X=x) \leftarrow \text{Optimal classifier on accuracy}$$

↳ min. error rate

$$P(Y=k|X=x) = \frac{P(Y=k) \cdot P(X=x|Y=k)}{\underbrace{P(X=x)}_{\text{Evidence}}}$$

$P(Y=k)$: Prior

$P(X=x|Y=k)$: Likelihood

↳ $d : 2^{d+1}$ distributions
[Curse of dimensionality]

Avoid Correlation across features.

Ensure independence across features.

$$\prod_{i=1}^d \Pr(X_i = x_i | Y=k)$$

↳ ~~Naive~~ - Bayes Classifier
Naive

7/3

$$\hat{y} \in \{1, 2, \dots, k\}$$

Bayes Classification $\rightarrow \operatorname{argmax}_{i \in [k]} P(y=i | x=\hat{x})$

↪ Optimal

Bayes Theorem:
$$P(A|B) = \frac{P(B|A) \cdot P(A)}{\sum_{\forall A} P(B|A) \cdot P(A)} = \frac{P(B|A) \cdot P(A)}{P(B)}$$

$$P(y=i | x=\hat{x}) = \frac{P(x=\hat{x} | y=i) \cdot P_0(y=i)}{P(x=\hat{x})} \leftarrow \operatorname{argmax}$$

$$= \frac{\text{Likelihood} \times \text{Prior}}{\text{Evidence}}$$

↪ does not play a role in decision

$$\Rightarrow \operatorname{argmax}_{i \in [k]} P(x=\hat{x} | y=i) \cdot P(y=i)$$

$$\Rightarrow \operatorname{argmax}_{i \in [k]} \prod_{j=1}^n P(x_j = \hat{x}_j | y=i) \cdot P(y=i)$$

2^{2^n} distributions - Curse of dimensionality

→ Decision Trees :

$$x = (x_1, x_2) \in \mathbb{R}^2$$

$$y \in \{0, 1\}$$

↙ Cricket
↘ Tennis

Information Gain :

Entropy :

2 messages → 1 bit

4 messages → 2 bits

$R_{Y|X}$ S/C

0.4 0 0

0.3 0 1

0.3 1 0

(1 1) ×

$$0.4 \times 2 + 0.3 \times 2 + 0.3$$

$$= 1.7$$

$$\text{Entropy} = - \sum_{i=1}^k p_i \cdot \log_2(p_i)$$

$$\left[\lim_{p \rightarrow 0} p \cdot \log(1/p) = 0 \right]$$

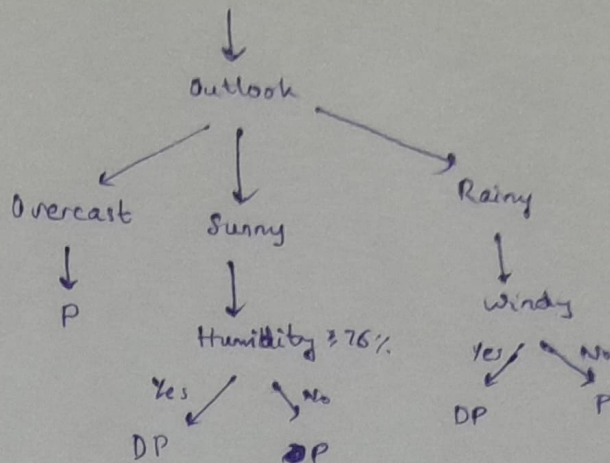
$$H(S) = \sum_{i \in [k]} \frac{S_i}{S} \times \log\left(\frac{S}{S_i}\right)$$

$$\text{Inf Gain} = \mathcal{IG}(S, A) = H(S) - H(S|A)$$

$$= H(S) - \sum_t \frac{S_t}{S} H(t)$$

$t \in T$ for diff. values present in S for Attribute A

Sort by descending order of $\mathcal{IG}$



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→ CART:

only 2 children

Gini Index: $\sum p_L(1-p_L)$

Pruning to avoid over-fitting

→ KNN:

$$D = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$$

$$|x_{\sigma(1)} - x| \leq |x_{\sigma(2)} - x| \leq \dots \leq |x_{\sigma(n)} - x|$$

$\hat{y}$: most frequently occurring label in $x_{\sigma(1)}, x_{\sigma(2)}, \dots, x_{\sigma(n)}$

Continuous features: Euclidean metric

Discrete features: Hamming distance

$$d: X \times X \rightarrow \mathbb{R}_+$$

$$(i) d(x, x) = 0$$

$$(ii) d(x, y) = 0 \text{ if } x = y \quad [d(x, y) = d(y, x)]$$

$$(iii) d(x, y) + d(y, z) \geq d(x, z)$$

As $n \rightarrow \infty$, 1-NN error rate is at most 2 times Bayes Classifier

Combining Classifiers

Bagging

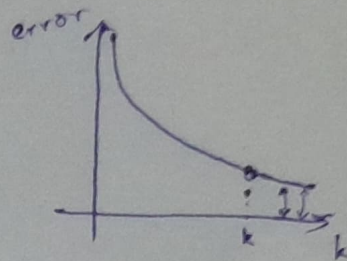
(Majority vote)

Boosting

(Weighted accuracy)

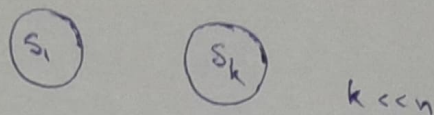
Decision Tree:

Small change in data leads to diff. tree



→ Clustering :

Hard clustering ← this course
Soft Clustering



$$S_i \cap S_j = \emptyset \quad [i \neq j]$$

$$\cup S_i = (x_1, x_2, \dots, x_n)$$

Connectivity based clustering

Agglomerative

(Bottom-Up)

~~AGNES~~ (AGNES)

Handles outliers

Divisive

~~Top-Bottom~~

(Top-Down)

(DIANA)

Outliers may disrupt

Standard Scalar : $\frac{z - \mu}{\sigma} \leftarrow z\text{-score}$

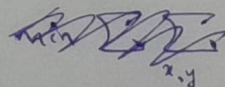
Centroid Based

Minimize variance

$$\min_{S_1, S_2, \dots, S_k} \sum_{i=1}^k |S_i| \cdot \text{Var } S_i$$

$$\min_{S_1, S_2, \dots, S_k} \sum_{i=1}^k \sum_{x \in S_i} \|x - \mu_i\|^2$$

$$\mu_i = \frac{1}{|S_i|} \sum_{x_j \in S_i} x_j$$



$$\min \sum_{i=1}^k \frac{1}{|S_i|} \sum_{x, y \in S_i} \|x - y\|^2$$

But All of them are NP-Hard.

Calculate Heuristically

K-Means:

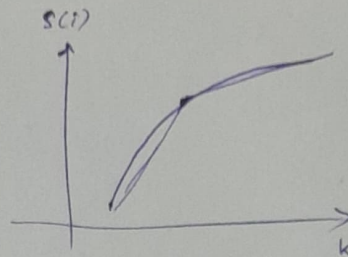
Randomly select k points as centers

Assign each point to the cluster
closest to the center

Compute mean of all points in the
cluster

Call these points the new centers

Repeat



Which k best?

(i) Project & Visualize

(ii) Silhouette Score [i : point]

$$S(i) = \frac{u(i) - a(i)}{\max\{a(i), b(i)\}}$$

$S(i) \uparrow \Rightarrow$ Better

(iii) Davies - Bouldin Index

$$DB = \frac{1}{k} \sum_i \max_j \frac{\Delta x_i + \Delta x_j}{\delta(x_i, x_j)}$$

Δ : Intracluster dist.

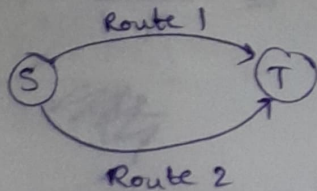
δ : Intercluster dist.

i : cluster

$$\frac{\sum S(i)}{n} \leftarrow \text{Best} [q, \text{for } k]$$

Minimize DB

Markov Decision Process (MDP) : $\langle S, P, R, A \rangle$



$$(H|H) = 0.6$$

$$(L|L) = 0.6$$

$$(L|H) = 0.4$$

$$(H|L) = 0.4$$

R_1	R_2
H	H
H	L
L	H
L	L

Value Iteration
Policy Iteration
Linear Programming

} $\Rightarrow$ Backward Induction

Monte Carlo Simulator

$\Downarrow$

Φ Learning

SARSA

DDPG

e.g. $|S| = 1$

$S = \{s\}$

$A = \{1, 2, \dots, k\}$

Value iteration when $R(s, a) = 1$ with prob. p_a & 0 otherwise

$$:= p_a$$

~~Value iteration~~

$t=T$

$|A| \times |S|$

$|A|^t$

$|A|^t$

$|A|^t$

Round Robin : 0.95 T

only 1 : 0.9 T

Expected reward : $\hat{r}_i$ for each arm

@ every t : select arm with highest $\hat{r}_i$.

$$V(s, a) = \max R(s, a) + \gamma V(s, a)$$

$$\Rightarrow V(s, a) = \frac{R(s, a)}{1 - \gamma} \Rightarrow V(s) = \max_a \frac{\mu_a}{1 - \gamma}$$

$$\frac{\mu_a}{1 - \gamma}$$

$$\begin{matrix} \mu_1 = 0.2 \\ \mu_2 = 0.3 \end{matrix} \left\{ \begin{array}{l} \text{Actions} \\ \text{Arms} \end{array} \right.$$

Multi Armed Bandits:

[Syllabus until UCB]

Greedy

ϵ -Greedy

UCB (Upper confidence bound)

Thompson Sampling

MAB: Classical Problem in decision-making under uncertainty

Key elements:

K arms (actions/choices)

Each arm $\rightarrow$ stochastic reward

At each round, select one arm and observe reward, observing: μ_a

Goal: Max. cumulative reward over T rounds

Learn while ~~learning~~ optimising
balance b/w exploring and exploitation

~~1, 0, 0~~

0.9

1

1, 0, 0

$\hat{\mu}_1 = \frac{1}{2}$

$0.9 \times 0.1 \times 0.1 \times 0.8 \times 0.8 \times 0.8$

0.8

2

1, 1, 1

$\hat{\mu}_2 = 1$

Round Robin: 0.85 T

only 1: 0.9 T

Expected reward: $\hat{\mu}_i$ for each arm

@ every t: select arm with highest $\hat{\mu}_i$.

Exploitation

$$\hat{\mu}_{a,t+1} = \frac{\hat{\mu}_a^t \times N_{a,t} + 1}{N_{a,t} + 1} \quad \text{if } R=1$$

$$= \frac{\hat{\mu}_a^t \times N_{a,t}}{N_{a,t} + 1} \quad \text{if } R=0$$

$$N_{\tilde{a},t+1} = N_{\tilde{a},t}$$

$$N_{a,t+1} = N_{a,t} + 1$$

$$\hat{\mu}_a^{t+1} = \hat{\mu}_a^t$$

$$N_{\tilde{a},t+1} = N_{\tilde{a},t}$$

$$\textcircled{1} \textcircled{4} \textcircled{3}$$

$$1, 0, 0$$

$$\textcircled{2} \textcircled{4} \textcircled{6}$$

$$1, 1, 1$$

$$\hat{\mu}_a^1 = 1$$

$$\hat{\mu}_b^2 = 1$$

$$\hat{\mu}_a^3 = 0.5$$

$$\hat{\mu}_b^3 = 1, \hat{\mu}_b^4 = 1$$

$$\hat{\mu}_a^6 = .13$$

$$\hat{\mu}_b^7 = .1$$

Greedy fails many times $\therefore$ Only exploitation

$\therefore$ Some exploration: ϵ -Greedy

(Exploration & Exploitation)

At each t time step:

Greedy with prob. $1-\epsilon$

choose arm with prob. ϵ

Concept of Regret:

Stochastic diff. b/w
optimal and achieved
performance

Regret $\downarrow \Rightarrow$ Better

$$\text{Reg}_A^{\text{expected}}(T) = E \max_{1,2,\dots,K} \left[\sum_{t=1}^T r_{i^*,t} - r_{i_t,t} \right]$$

$$\text{Reg}_A^{\text{pseudo}}(T) = \sum_{t=1}^T \max_{1,2,\dots,K} E[r_i] - E \left[\sum_{t=1}^T r_{i_t,t} \right]$$

$$= \sum_{t=1}^T E[\mu_{i^*} - \mu_{A_t}]$$

$$\text{Reg}_A(T) = T \times \mu_{i^*} - \sum_{t=1}^T r_{i_t,t}$$

Algorithm A

$r_{i_t,t}$: reward observed

Atleast $\epsilon \times \frac{k-1}{k} \times T$

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MDP - Puttberman

MAB - Bubeck & Cesa-branc

Multi-armed Bandits

Decision

$\{1, 2, \dots, k\} \mu_i$

0.9

0.6

11100

11110

0.6

0.8

$$T \mu_{i,t} - \sum_{t=1}^T r_{i,t} \leftarrow \underbrace{\epsilon \left(\frac{k-1}{k} \right) T}_{o(T)}$$

$\therefore \epsilon$ should keep changing

$\hat{\mu}_i (0.89 \ 0.91) \ 0.59 \ 0.61$

$$\boxed{\epsilon = 1/t}$$

$$\mu_i \in (\hat{\mu}_t - c_{i,t}, \hat{\mu}_t + c_{i,t})$$

Confidence Bound $\rightarrow$ guide exploration

• Upper Confidence Bound / Reward Estimate

$$\hat{\mu}_i^+ = \hat{\mu}_i + c_{i,t}$$

Select arm with highest UCB.

Algo for UCB1

$$\sqrt{\frac{\alpha \ln T}{N_{i,t}}} \quad \alpha: \text{fixed } (2/3, \dots)$$

input: $\alpha > 0$

for $t \in \{1, 2, \dots, K\}$ do

select an arm $i_t \leftarrow t$

Observe reward $r_{i_t, t}$

Set initial $\hat{\mu}_i \leftarrow r_{i_t, t}$

end for

for $k \in \{K+1, \dots, T\}$ do

Select an arm $i_k \in \arg\max_{i \in [K]} \left(\hat{\mu}_{i, k} + \sqrt{\frac{\alpha \ln T}{N_{i, k}}} \right)$

observe reward $r_{i_k, k}$

Update $\hat{\mu}_{i_k}$

end for

~~Reg~~ (i) Underestimating of optimal

$$\hat{\mu}_{i, k} < \mu^* - c_{t, s}$$

(ii) Overestimation of suboptimal

$$\hat{\mu}_{i, k} \geq \mu_i + c_{t, s}$$

(iii) High uncertainty

$$\mu^* \leq \mu_i + 2c_{t, s}$$

s: no. of times
pulled

$$\text{Reg}_{\text{UCB1}}(T) \leq \left(8 \sum_{i: \mu_i < \mu^*} \frac{\ln T}{\Delta_i} \right) + \left(1 + \frac{\pi^2}{3} \right) \left(\sum_{j=1}^K \Delta_j \right)$$

$$\Delta_i = \mu^* - \mu_i$$

Regret: $O(\ln T)$

$\forall K > 1$

Support $\in [0, 1]$

Worst case: $\Delta_i = \frac{1}{\sqrt{T}}$

$$O(\sqrt{T} \cdot \ln T)$$

Pseudo regret:

$$T \times \mu_i^* - \sum_{t=1}^T \mu_{i_t, t}$$

$$i \neq i^*: \text{Regret} = \Delta_i \times E[N_{i, t}]$$

When: [Suboptimal part]

$$\hat{\mu}_i^+ \geq \hat{\mu}_{i^*}^+$$

$$c_{t, s} = \sqrt{\frac{2 \ln t}{s}}$$

$$\left. \begin{aligned} P(\text{Event 1}) &\leq t^{-4} \\ P(\text{Event 2}) &\leq t^{-4} \end{aligned} \right\} \begin{array}{l} \text{These events} \\ \text{contribute } o(1) \\ \text{to the exp.} \\ \text{regret} \end{array}$$

$$\sum_{t=1}^T 2t^{-4} \rightarrow \frac{\pi^2}{3}$$

Chernoff - Hoeffdings bounds

Bounding Events (1 & 2)

[Concentration] ~~Concentration~~

Bounding Event 3 :

$$\sqrt{\frac{2 \ln T}{N_{i,t}}} > \frac{\Delta_i}{2} \Rightarrow N_{i,t} < \frac{8 \ln T}{\Delta_i^2}$$

Haven't pulled ith arm enough

$$\mathbb{E}[N_{i,t}] \leq \frac{8 \ln T}{\Delta_i^2} + \left(1 + \frac{\pi^2}{3}\right)$$

Additive Bandits $\rightarrow$ UCB 1

$$\mathbb{E} N_{i,t} < T$$

$$f: S \leq 2^k \rightarrow \mathbb{R}$$

$$\mu_i \sim \text{Gamma}(\alpha, \beta) \\ o(\ln T)$$

$$\mu_i \sim T(\alpha_i, \beta_i)$$

$\rightarrow$ Bayesian Approach / Thomson Sampling
(Not part of Syllabus)

$\rightarrow$ Variations of Multi-armed Bandits :

1. Stochastic - Rewards drawn from i.i.d

Adversarial - Rewards chosen
by Adversary
(EXP3)

2. Stationary

Non stationary

3. Volatile / sleeping Bandits $\rightarrow$ Unavailable

4. Budgeting Bandits - arm pull with
associated costs
max. reward

5. ~~Competitive Bandits~~
Competitive Bandits